(i)

$$y = x^4 - 6x^2 + 9 = (x^2 - 3)^2$$

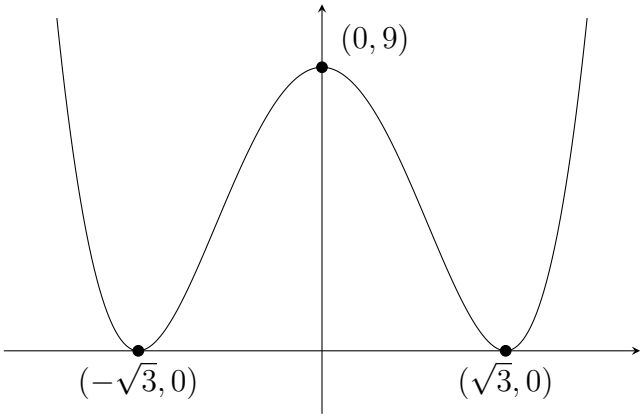
$$\frac{dy}{dx} = 4x^3 - 12x = 4x(x^2 - 3)$$

$$\frac{dy}{dx} = 0 \implies x = 0,\ \pm\sqrt{3}$$

$$y(0) = 9, \qquad y(\pm\sqrt{3}) = 0$$

Stationary points are

$$(-\sqrt{3}, 0), \quad (0, 9), \quad (\sqrt{3}, 0)$$



$$x^4 - 6x^2 + b = 0 \iff x^4 - 6x^2 = -b$$

$$x^4 - 6x^2 = (x^2 - 3)^2 - 9$$

n	0	1	2	3	4
b	$b > 9$	none	$b < 0$ or $b = 9$	$b = 0$	$0 < b < 9$

(ii)

$$\frac{dy}{dx} = 4x^3 - 12x + a, \qquad \frac{d^2y}{dx^2} = 12x^2 - 12$$

$$\frac{d^2y}{dx^2} = 0 \implies x = \pm 1$$

When $x = 1,\ 4 - 12 + a = 0 \implies a = 8$

When $x = -1,\ -4 + 12 + a = 0 \implies a = -8$
When $a = 8$,

$$y = x^4 - 6x^2 + 8x + b, \qquad \frac{dy}{dx} = 4(x-1)^2(x+2)$$

$$\frac{dy}{dx} < 0 \quad (x < -2), \qquad \frac{dy}{dx} \geq 0 \quad (x > -2)$$

Minimum point has $x = -2,\ y(-2) = b - 24$

b	$b > 24$	$b = 24$	$b < 24$
roots	0	1	2

$\because$ When $a = -8,\ x^4 - 6x^2 - 8x + b = f_8(-x)$
$\therefore$ The roots count is the same for $a = -8$

(iii)

$$y = x^4 - 6x^2 + ax, \qquad a > 8$$

$$\frac{dy}{dx} = 4x^3 - 12x + a$$

Let $g(x) = 4x^3 - 12x + a$

$$g'(x) = 12x^2 - 12$$

$$g(-1) = a + 8 > 0, \qquad g(1) = a - 8 > 0$$

$$g(-2) = a - 8 > 0, \qquad \lim_{x \to -\infty} g(x) = -\infty$$

There is one stationary point at $x = \alpha$, where $\alpha < -2$

$$4\alpha^3 - 12\alpha + a = 0$$

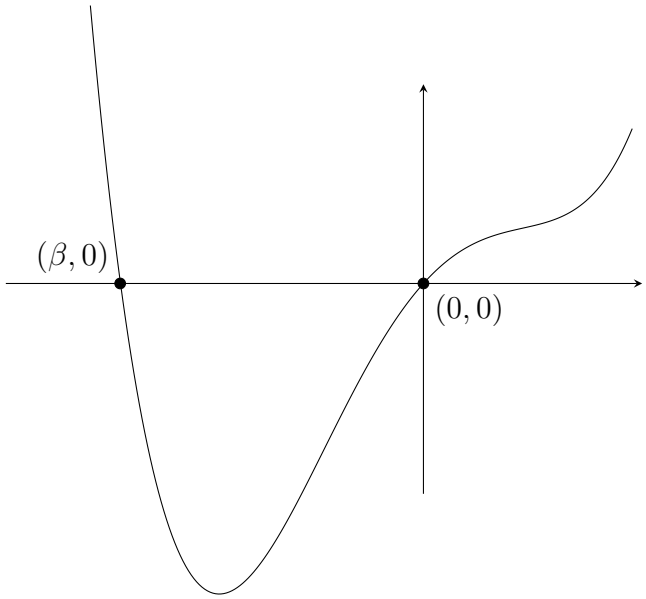
$$a = 12\alpha - 4\alpha^3$$

$$y(\alpha) = \alpha^4 - 6\alpha^2 + a\alpha = 3\alpha^2(2 - \alpha^2) < 0$$

$$y = x(x^3 - 6x + a)$$

Let β be the negative root of $x^3 - 6x + a = 0$

$$\beta < \alpha < -2, \qquad y = 0 \text{ at } x = \beta, 0$$



This solution was submitted by **Jon**
Published and refined by **Jon**

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